Girsanov and Risk-Neutral Pricing

Changing probabilities, moving drifts, and pricing by expectation

Lecture 6

One question drives the whole lecture

The question

Can we change probabilities so that the same risky asset has drift r instead of drift μ?

1. Reweight scenarios and derive Bayes' rule.
2. See Gaussian tilting move a mean.
3. Extend the idea to Brownian paths.
4. Build the stochastic exponential and state Girsanov.
5. Apply the theorem to discounted stock prices.
6. Recover pricing by conditional expectation.

Reweighting Scenarios and Bayes' Rule

A Gaussian tail is invisible to direct Monte Carlo

Let $X \sim \mathcal{N}(0,1)$ under $\mathbb{P}$. We want

$$p = \mathbb{P}(X > 5) = 1 - \Phi(5) \approx 2.87 \times 10^{-7}.$$

With $n = 10^6$ independent draws, direct Monte Carlo has

$$\widehat{p}_{\mathrm{MC}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X_i > 5\}}, \qquad \mathbb{E}^{\mathbb{P}}[\text{number of hits}] = np \approx 0.29.$$

Too few tail observations

The chance of seeing no hit is $(1-p)^n \approx 75\%$; then the estimate is zero.

Change the sampling probabilities

Give more weight to the tail: under a new law $\mathbb{Q}$, let $X \sim \mathcal{N}(5,1)$, so $\mathbb{Q}(X > 5) = 1/2$. We will correct for the change of probabilities after sampling.

On a finite space a density is a probability ratio

Let $\Omega = \{\omega_1, \ldots, \omega_m\}$, with positive probabilities under $\mathbb{P}$ and $\mathbb{Q}$. Define

$$Z(\omega_i) = \frac{\mathbb{Q}(\omega_i)}{\mathbb{P}(\omega_i)}.$$

Then

$$Z > 0, \qquad \mathbb{E}^{\mathbb{P}}[Z] = 1, \qquad \boxed{\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[ZY].}$$

Interpretation

$Z > 1$ upweights a scenario; $Z < 1$ downweights it.

A three-state regime

	ω_1	ω_2	ω_3
$\mathbb{P}$	$1/2$	$1/3$	$1/6$
$\mathbb{Q}$	$1/4$	$1/2$	$1/4$

YOUR TURN Compute Z, identify which states are upweighted, and check $\mathbb{E}^{\mathbb{P}}[Z] = 1$.

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YOUR TURN Compute Z, identify which states are upweighted, and check $\mathbb{E}^{\mathbb{P}}[Z] = 1$.

$$\boxed{Z(\omega_1) = \tfrac{1}{2}, \quad Z(\omega_2) = \tfrac{3}{2}, \quad Z(\omega_3) = \tfrac{3}{2}.}$$

Use the density rather than recomputing every probability

In the previous model, let $Y = (-2, 1, 4)$.

YOUR TURN Compute $\mathbb{E}^{\mathbb{Q}}[Y]$ directly, then as $\mathbb{E}^{\mathbb{P}}[ZY]$.

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$$\mathbb{E}^{\mathbb{Q}}[Y] = -\frac{1}{2} + \frac{1}{2} + 1 = 1,$$

$$\mathbb{E}^{\mathbb{P}}[ZY] = \frac{1}{2}\frac{1}{2}(-2) + \frac{1}{3}\frac{3}{2}(1) + \frac{1}{6}\frac{3}{2}(4) = 1.$$

The general construction needs positivity and unit mean

Take $Z \geq 0$ with $\mathbb{E}^{\mathbb{P}}[Z] = 1$, and define

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A].$$

Then $Z = \mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ and

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[ZY].}$$

Equivalence

If $Z > 0$ almost surely, $\mathbb{P}$ and $\mathbb{Q}$ have the same null events. No possible scenario is created or deleted.

Conditional expectations need conditional normalization

For $\mathcal{G} \subseteq \mathcal{F}$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}] = \frac{\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]}{\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]}.}$$

Why the denominator appears

It renormalizes the weights after the information in $\mathcal{G}$ has been revealed.

Prove Bayes' rule from its defining test

Set

$$X = \frac{\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]}{\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]}.$$

YOUR TURN For $A \in \mathcal{G}$, show that $\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A X] = \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A Y]$.

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$$\mathbb{E}^{\mathbb{P}}[\mathbf{1}_A X \mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]] = \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A \mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]] = \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A ZY].$$

Known information is unchanged by reweighting

If Y is $\mathcal{G}$-measurable, then

$$\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}] = Y\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}],$$

so

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}] = Y.}$$

YOUR TURN Recover the unconditional identity by taking the trivial sigma-field.

Gaussian Tilting Moves a Mean

The one-dimensional prototype fixes the sign convention

Let $X \sim \mathcal{N}(0,1)$ under $\mathbb{P}$ and

$$Z = \exp\left(-\theta X - \frac{1}{2}\theta^2\right).$$

The moment-generating function (MGF) is $M_X(u) = \mathbb{E}^{\mathbb{P}}[e^{uX}] = e^{u^2/2}$. Hence

$$\mathbb{E}^{\mathbb{P}}[Z] = e^{-\theta^2/2} M_X(-\theta) = 1.$$

YOUR TURN For $\theta > 0$, does Z favor positive or negative X?

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YOUR TURN For $\theta > 0$, does Z favor positive or negative X?

It favors negative values because the weight contains $e^{-\theta X}$.

Identifies the new law

$$e^{-\theta x-\theta^2/2}\frac{e^{-x^2/2}}{\sqrt{2\pi}}=\frac{1}{\sqrt{2\pi}}e^{-(x+\theta)^2/2}.$$

Therefore

$$\boxed{X\sim\mathcal{N}(-\theta,1)\text{ under }\mathbb{Q}.}$$

Prototype of Girsanov

The mean changes; the variance stays equal to one.

Reverse-engineer a target Gaussian mean

Suppose $X \sim \mathcal{N}(0, \sigma^2)$ under $\mathbb{P}$.

YOUR TURN Find $Z = e^{aX+b}$ that makes $X \sim \mathcal{N}(m, \sigma^2)$ under $\mathbb{Q}$.

Reverse-engineer a target Gaussian mean

Suppose $X \sim \mathcal{N}(0, \sigma^2)$ under $\mathbb{P}$.

YOUR TURN Find $Z = e^{aX+b}$ that makes $X \sim \mathcal{N}(m, \sigma^2)$ under $\mathbb{Q}$.

$$\boxed{a = \frac{m}{\sigma^2}, \qquad b = -\frac{m^2}{2\sigma^2}.}$$

The term b enforces $\mathbb{E}^{\mathbb{P}}[Z] = 1$.

Reweighting and translation give the same law identity

With $Z = e^{-\theta X - \theta^2/2}$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[f(X)] = \mathbb{E}^{\mathbb{P}}[Zf(X)] = \mathbb{E}^{\mathbb{P}}[f(X-\theta)].}$$

YOUR TURN Which equality is the definition of $\mathbb{Q}$, and which equality uses the Gaussian law?

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YOUR TURN Which equality is the definition of $\mathbb{Q}$, and which equality uses the Gaussian law?

The first is change of measure; the second is a distributional consequence.

A shifted tail needs a likelihood correction

Recall $p = \mathbb{P}(X > 5) \approx 2.87 \times 10^{-7}$. Sample $X_1, \dots, X_n$ under $\mathbb{Q}$, where $X \sim \mathcal{N}(5, 1)$.

Shift the mean only

$$\widehat{p}_{\text{shift}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X_i > 5\}} \longrightarrow \mathbb{Q}(X > 5) = \frac{1}{2}.$$

This estimates the tail under $\mathbb{Q}$, not the original tail.

Restore the original weights

$$w(x) = \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}(x) = e^{-5x + 25/2},$$

$$\widehat{p}_{\text{IS}} = \frac{1}{n} \sum_{i=1}^{n} w(X_i) \mathbf{1}_{\{X_i > 5\}}.$$

$$\boxed{\mathbb{E}^{\mathbb{Q}}[w(X)\mathbf{1}_{\{X>5\}}] = \mathbb{P}(X > 5) = p.}$$

Why Change the Probability Measure?

The payoff stays the same while scenario weights change

For a terminal payoff H, valuation depends on its possible values and on the probabilities assigned to the scenarios.

Change of viewpoint

A change of measure does not modify the state, the stock path, or $H(\omega)$. It changes how strongly each state is weighted.

What Girsanov will do

The drift may change. The diffusion coefficient will not.

A pricing measure is not a forecast

MODEL ASSUMPTION Under $\mathbb{P}$, dynamics describe statistical behavior:

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t^{\mathbb{P}}.$$

MARKET ASSUMPTION Under $\mathbb{Q}$, discounted traded prices should be martingales:

$$\frac{\mathrm{d}S_t}{S_t} = r_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t^{\mathbb{Q}}.$$

Two practical reasons to change the measure

MARKET ASSUMPTION **Option pricing.** Reweight paths so discounted traded prices have zero drift. Then a price is a discounted expectation under $\mathbb{Q}$.

Rare-event simulation

Reweight paths toward a rare loss (or barrier hit), estimate it more often, then correct the estimate with the likelihood ratio $\mathrm{d}\mathbb{P}/\,\mathrm{d}\mathbb{Q}$.

Same paths, different probabilities

The underlying scenarios and payoff do not change; only their weights do. The new measure is chosen to make the calculation or simulation easier.

A measure change is more than changing one variable's law

Tilt one variable

For X, a density proportional to $e^{\theta X}$ changes the marginal distribution of X. It says nothing by itself about how other variables or the rest of the path are reweighted.

Change the measure

Choose a positive random weight Z_T with $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$, and set $\mathrm{d}\mathbb{Q} = Z_T \, \mathrm{d}\mathbb{P}$. Every event and every payoff is reweighted:

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z_T \mathbf{1}_A], \qquad \mathbb{E}^{\mathbb{Q}}[H] = \mathbb{E}^{\mathbb{P}}[Z_T H].$$

An exponential tilt of X *can* define a measure change; Girsanov's density extends that idea to entire Brownian paths.

Discounting reveals the drift to remove

Suppose $B_t = e^{rt}$ and

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

YOUR TURN Compute the dynamics of $\bar{S}_t = S_t/B_t$. Which term must disappear?

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YOUR TURN Compute the dynamics of $\bar{S}_t = S_t/B_t$. Which term must disappear?

$$\boxed{\frac{\mathrm{d}\bar{S}_t}{\bar{S}_t} = (\mu - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.}$$

The ratio $(\mu - r)/\sigma$ is excess drift measured in volatility units.

From Gaussian Tilting to Brownian Paths

A deterministic integrand produces one Gaussian variable

Let $h \in L^2([0,T])$ be deterministic and define

$$X = \int_0^T h_s \, \mathrm{d}W_s \sim \mathcal{N}\left(0, \int_0^T h_s^2 \, \mathrm{d}s\right).$$

Set $v = \int_0^T h_s^2 \, \mathrm{d}s$. The Gaussian MGF says $\mathbb{E}[e^X] = e^{v/2}$.

YOUR TURN Find a unit-mean exponential weight built from X, and write it as a density for reweighting Brownian paths.

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The normalizing correction is $-v/2$, so the path density is

$$\boxed{Z_T = \exp\left(\int_0^T h_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^T h_s^2 \, \mathrm{d}s\right).}$$

Cameron–Martin absorbs deterministic drift

Define $\mathbb{Q}$ by $\mathrm{d}\mathbb{Q} = Z_T \, \mathrm{d}\mathbb{P}$. Then

$$\boxed{B_t = W_t - \int_0^t h_s \, \mathrm{d}s \quad \text{is Brownian under } \mathbb{Q}.}$$

Equivalently,

$$\mathrm{d}W_t = \mathrm{d}B_t + h_t \, \mathrm{d}t.$$

Pathwise interpretation

The density rewards paths aligned with h, so W acquires drift $\int h$ under $\mathbb{Q}$.

A constant drift checks the theorem against Gaussian tilting

Let $h_t \equiv \theta$. Then

$$Z_T = e^{\theta W_T - \theta^2 T/2}, \qquad B_t = W_t - \theta t.$$

YOUR TURN What is the law of W_T under $\mathbb{Q}$?

A constant drift checks the theorem against Gaussian tilting

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$$Z_T = e^{\theta W_T - \theta^2 T/2}, \qquad B_t = W_t - \theta t.$$

YOUR TURN What is the law of W_T under $\mathbb{Q}$?

Since $W_T = B_T + \theta T$,

$$\boxed{W_T \sim \mathcal{N}(\theta T, T) \text{ under } \mathbb{Q}.}$$

Random adapted drift requires a new argument

For deterministic h, $\int h \, \mathrm{d}W$ is Gaussian, so one mgf normalizes the density.

What changes

If h is replaced by a random adapted process ϕ, the stochastic integral need not be Gaussian. We must use stochastic calculus to control $\mathbb{E}[Z_T]$.

The Stochastic Exponential

Adapted drift leads to an exponential process

Let ϕ be progressively measurable and suppose

$$\int_0^T \phi_s^2 \, \mathrm{d}s < \infty \quad \text{a.s.}$$

Define

$$Z_t = \exp\left(\int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2} \int_0^t \phi_s^2 \, \mathrm{d}s \right).$$

YOUR TURN Compute dZ_t before calculating.

Itô's formula explains the compensation

Set

$$Y_t = \int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2} \int_0^t \phi_s^2 \, \mathrm{d}s.$$

Then

$$\mathrm{d}Y_t = \phi_t \, \mathrm{d}W_t - \frac{1}{2}\phi_t^2 \, \mathrm{d}t, \qquad \mathrm{d}[Y]_t = \phi_t^2 \, \mathrm{d}t.$$

Applying Itô to $Z_t = e^{Y_t}$,

$$\mathrm{d}Z_t = Z_t \, \mathrm{d}Y_t + \frac{1}{2} Z_t \, \mathrm{d}[Y]_t.$$

Quadratic variation cancels the finite-variation drift

$$\begin{aligned} \mathrm{d}Z_t &= Z_t \left(\phi_t \, \mathrm{d}W_t - \frac{1}{2}\phi_t^2 \, \mathrm{d}t \right) + \frac{1}{2} Z_t \phi_t^2 \, \mathrm{d}t \\ &= \boxed{Z_t \phi_t \, \mathrm{d}W_t}. \end{aligned}$$

YOUR TURN Does zero drift already prove that Z is a true martingale?

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YOUR TURN Does zero drift already prove that Z is a true martingale?

No. It first gives a local martingale; integrability is still required.

Novikov is a usable sufficient condition

Novikov condition

If

$$\boxed{\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \phi_s^2\,\mathrm{d}s\right)\right] < \infty,}$$

then Z is a true martingale and $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$.

We use the condition; its proof is beyond today's scope.

Bounded drift makes Novikov immediate

If $|\phi_t| \le C$, then

$$\int_0^T \phi_s^2 \, \mathrm{d}s \le C^2 T$$

and

$$\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \phi_s^2 \, \mathrm{d}s} \right] \le e^{C^2 T/2} < \infty.$$

YOUR TURN Does finite energy almost surely imply Novikov automatically?

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No. Almost-sure finiteness does not control an exponential expectation.

Diagnose three candidate densities

candidate	information
$e^{\theta W_T - \theta^2 T/2}$	$\theta \in \mathbb{R}$
$e^{\int_0^T \phi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^T \phi_s^2 \, \mathrm{d}s}$	$\lvert\phi_s\rvert \leq C$
$e^{\int_0^T \psi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^T \psi_s^2 \, \mathrm{d}s}$	$\int_0^T \psi_s^2 \, \mathrm{d}s < \infty$ a.s.

YOUR TURN Which can immediately define a new probability measure? Justify.

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YOUR TURN Which can immediately define a new probability measure? Justify.

The first is Gaussian, the second satisfies Novikov, and the third needs more information.

Girsanov Moves Drift into the Measure

Girsanov identifies the new Brownian motion

Assume Z is a true martingale and

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T, \qquad Z_t = e^{\int_0^t \phi_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2 \,\mathrm{d}s}.$$

Girsanov theorem

Under $\mathbb{Q}$,

$$\boxed{B_t = W_t - \int_0^t \phi_s \,\mathrm{d}s}$$

is Brownian motion.

Write the differential substitution before using the theorem

The measure is defined by the full density process

$$Z_t = \exp\left(\int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2 \, \mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Under $\mathbb{Q}$,

$$B_t = W_t - \int_0^t \phi_s \, \mathrm{d}s \quad \Longleftrightarrow \quad \boxed{\mathrm{d}W_t = \mathrm{d}B_t + \phi_t \, \mathrm{d}t.}$$

CHECK THE SIGN The density sign fixes the Brownian shift. State the theorem and solve for $\mathrm{d}W_t$ before substituting into an SDE.

A generic Itô process keeps its volatility

Under $\mathbb{P}$, suppose

$$\mathrm{d}X_t = b_t\,\mathrm{d}t + a_t\,\mathrm{d}W_t.$$

Use the density

$$Z_t = \exp\left(\int_0^t \phi_s\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2\,\mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Then under $\mathbb{Q}$,

$$\boxed{\mathrm{d}X_t = (b_t + a_t\phi_t)\,\mathrm{d}t + a_t\,\mathrm{d}B_t.}$$

Invariant

Only the drift changes. The diffusion coefficient remains a_t.

Choose the shift to remove a target drift

Let $a_t \neq 0$ in

$$\mathrm{d}X_t = b_t\,\mathrm{d}t + a_t\,\mathrm{d}W_t.$$

YOUR TURN Choose ϕ_t so that X has zero drift under $\mathbb{Q}$.

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$$\boxed{\phi_t = -\frac{b_t}{a_t}.}$$

The corresponding density is

$$Z_t = \exp\left(-\int_0^t \frac{b_s}{a_s}\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \frac{b_s^2}{a_s^2}\,\mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

This is a formal candidate until Z is shown to be a true martingale.

The theorem becomes useful through three translations

Girsanov has identified a new Brownian motion. We now use it to answer concrete questions.

1. Read the Brownian shift from the density $Z_t = \exp\big(\int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2 \, \mathrm{d}s\big)$.
2. Substitute $\mathrm{d}W$ and write the full SDE under $\mathbb{Q}$.
3. Use the new SDE to identify a law or compute an expectation.

The progression

We start with W, then an arithmetic diffusion, an adapted drift, and finally a geometric asset price.

A constant density moves the Brownian mean

Define

$$Z_T = e^{-\lambda W_T - \lambda^2 T/2}.$$

YOUR TURN Find the $\mathbb{Q}$-Brownian motion and the law of W_T under $\mathbb{Q}$.

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Define

$$Z_T = e^{-\lambda W_T - \lambda^2 T/2}.$$

YOUR TURN Find the $\mathbb{Q}$-Brownian motion and the law of W_T under $\mathbb{Q}$.

Here $\phi = -\lambda$, so

$$W_t^{\mathbb{Q}} = W_t + \lambda t, \qquad W_T = W_T^{\mathbb{Q}} - \lambda T,$$

and therefore

$$\boxed{W_T \sim \mathcal{N}(-\lambda T, T) \text{ under } \mathbb{Q}.}$$

The same shift removes drift from an arithmetic diffusion

Under $\mathbb{P}$, let

$$\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \sigma > 0,$$

and set $\lambda = \mu/\sigma$.

$$Z_T = e^{-\lambda W_T - \lambda^2 T/2}, \qquad W_t^{\mathbb{Q}} = W_t + \lambda t.$$

Since $\mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} - \lambda\,\mathrm{d}t$,

$$\boxed{\mathrm{d}X_t = \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

If $X_0 = x$, then under $\mathbb{Q}$,

$$\boxed{X_t \sim \mathcal{N}(x, \sigma^2 t).}$$

A change of measure can target any constant drift

Suppose again

$$\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t,$$

but now we want drift ν, not necessarily zero.

YOUR TURN Which Brownian shift produces $\mathrm{d}X_t = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}$?

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YOUR TURN Which Brownian shift produces $\mathrm{d}X_t = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}$?

Choose

$$\lambda = \frac{\mu - \nu}{\sigma}, \qquad Z_t = e^{-\lambda W_t - \lambda^2 t/2}, \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Then $W_t^{\mathbb{Q}} = W_t + \lambda t$ and

$$\boxed{\mathrm{d}X_t = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

A bounded adapted drift can also be absorbed

Under $\mathbb{P}$, consider

$$\mathrm{d}X_t = \sin(W_t)\,\mathrm{d}t + \mathrm{d}W_t, \qquad X_0 = 0.$$

Take $\phi_t = -\sin(W_t)$ and define $\mathbb{Q}$ by

$$Z_t = \exp\!\left(-\int_0^t \sin(W_s)\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \sin^2(W_s)\,\mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Because $|\phi_t| \le 1$, Novikov holds. Under $\mathbb{Q}$,

$$B_t = W_t + \int_0^t \sin(W_s)\,\mathrm{d}s$$

is Brownian, and substitution gives

$$\boxed{\mathrm{d}X_t = \mathrm{d}B_t.}$$

A geometric diffusion prepares the pricing measure

Under $\mathbb{P}$, suppose

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \sigma > 0.$$

To target a new drift ν, choose

$$\lambda = \frac{\mu - \nu}{\sigma}, \qquad Z_t = e^{-\lambda W_t - \lambda^2 t/2}, \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Then $W_t^{\mathbb{Q}} = W_t + \lambda t$ and

$$\boxed{\frac{\mathrm{d}S_t}{S_t} = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

MARKET ASSUMPTION For pricing with the bank account as numeraire, the required target is $\nu = r$. This is the next step.

New dynamics give laws; the density gives conditioning

Once the SDE is written under $\mathbb{Q}$, we can work directly with its new law. To translate a conditional expectation back to $\mathbb{P}$, use

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_s] = \frac{1}{Z_s}\mathbb{E}^{\mathbb{P}}[Z_T Y \mid \mathcal{F}_s], \qquad Z_s = \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_s].}$$

For the geometric example with constant ν, σ,

$$\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_s] = S_s e^{\nu(T-s)} = \frac{1}{Z_s}\mathbb{E}^{\mathbb{P}}[Z_T S_T \mid \mathcal{F}_s].$$

The new SDE gives the middle expression; the density translates it back to the historical measure.

The examples reveal one reusable workflow

1. **Target:** identify the drift wanted under the new measure.
2. **Density:** write Z_T with the correct sign and compensator.
3. **Validity:** prove or assume that Z is a true martingale.
4. **Substitution:** identify $W^{\mathbb{Q}}$, solve for $\mathrm{d}W$, and write the SDE under $\mathbb{Q}$.
5. **Use:** compute the new law or a conditional expectation.

YOUR TURN Which step remains unresolved if Novikov is unavailable?

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YOUR TURN Which step remains unresolved if Novikov is unavailable?

Only validity; the algebraic drift candidate still exists.

The Risk-Neutral Measure in Black–Scholes

Discounting exposes the drift that must be removed

Under $\mathbb{P}$,

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \qquad B_t^{(0)} = e^{\int_0^t r_s\,\mathrm{d}s}, \qquad D_t = (B_t^{(0)})^{-1}.$$

Since D has finite variation,

$$\boxed{\frac{\mathrm{d}(D_tS_t)}{D_tS_t} = (\mu_t - r_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t.}$$

The market price of risk is the required shift

Define

$$\lambda_t = \frac{\mu_t - r_t}{\sigma_t}.$$

Then

$$\frac{\mathrm{d}(D_t S_t)}{D_t S_t} = \lambda_t \sigma_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t.$$

YOUR TURN Should the density contain $+\int \lambda \,\mathrm{d}W$ or $-\int \lambda \,\mathrm{d}W$ to eliminate the drift?

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YOUR TURN Should the density contain $+\int \lambda \,\mathrm{d}W$ or $-\int \lambda \,\mathrm{d}W$ to eliminate the drift?

It must use the negative sign.

The pricing density has the negative market price of risk

$$\boxed{Z_t = \exp\left(-\int_0^t \lambda_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \lambda_s^2 \, \mathrm{d}s\right).}$$

Assume Z is a true martingale and define

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Girsanov gives

$$\boxed{W_t^{\mathbb{Q}} = W_t + \int_0^t \lambda_s \, \mathrm{d}s.}$$

Substitution replaces μ by r

$$\mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \,\mathrm{d}t.$$

Therefore

$$\begin{aligned}\frac{\mathrm{d}S_t}{S_t} &= \mu_t \,\mathrm{d}t + \sigma_t(\,\mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \,\mathrm{d}t) \\ &= (\mu_t - \sigma_t\lambda_t)\,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t^{\mathbb{Q}} \\ &= \boxed{r_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t^{\mathbb{Q}}}.\end{aligned}$$

Discounted stock prices lose their drift under $\mathbb{Q}$

Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = r_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Thus

$$\boxed{\mathrm{d}(D_tS_t) = \sigma_t D_t S_t\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

Conclusion

The discounted stock is a $\mathbb{Q}$-local martingale and, under suitable integrability, a true martingale.

Risk-neutral does not describe investor preferences

Correct meaning

$\mathbb{Q}$ is an equivalent measure under which discounted traded prices are martingales.

Incorrect meaning

$\mathbb{Q}$ does not claim that real investors expect every asset to grow at the risk-free rate.

It is a pricing representation/ a tool enforced/given by no arbitrage.

Split the call payoff into two probability questions

Under the risk-neutral measure,

$$S_T = s\exp\big((r - \tfrac{1}{2}\sigma^2)\tau + \sigma W_\tau^{\mathbb{Q}}\big), \qquad C = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

Write the payoff as $(S_T - K)\mathbf{1}_{\{S_T > K\}}$. Then

$$C = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T>K\}}] - Ke^{-r\tau}\mathbb{Q}(S_T > K).$$

YOUR TURN Using the lognormal formula, show that $\mathbb{Q}(S_T > K) = N(d_-)$, where $d_- = \frac{\log(s/K)+(r-\frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$.

Girsanov interprets the stock-weighted tail

Define the stock-numeraire measure by

$$\left.\frac{\mathrm{d}\mathbb{Q}^S}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = \frac{e^{-r\tau}S_T}{s} = \exp\left(\sigma W_\tau^{\mathbb{Q}} - \tfrac{1}{2}\sigma^2\tau\right).$$

This is a unit-mean density. By Girsanov,

$$W_u^S = W_u^{\mathbb{Q}} - \sigma u \quad \text{is Brownian under } \mathbb{Q}^S.$$

Thus the first payoff contribution becomes

$$e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T>K\}}] = s\,\mathbb{Q}^S(S_T > K).$$

Under $\mathbb{Q}^S$, the same event $S_T > K$ has probability $N(d_+)$, with

$$d_+ = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

The call formula is a risk-neutral Gaussian expectation

For $\tau = T - t$,

$$C(t,s) = s\,N(d_+) - Ke^{-r\tau}N(d_-),$$

$$d_\pm = \frac{\log(s/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_- = d_+ - \sigma\sqrt{\tau}.$$

Two measures, same exercise event

$N(d_-) = \mathbb{Q}(S_T > K)$ under the money-market measure, while $N(d_+) = \mathbb{Q}^S(S_T > K)$ under the stock-numeraire measure. Thus $sN(d_+)$ is the stock-weighted tail, and $Ke^{-r\tau}N(d_-)$ is the discounted strike tail.

Delta turns a pricing formula into a trading rule

$$\boxed{\Delta_t = C_s(t, S_t) = N(d_+(t, S_t)).}$$

YOUR TURN As maturity approaches, what should delta do when the call is deeply in the money? Deeply out of the money? Near the strike?

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$$\Delta_t \longrightarrow \begin{cases} 1, & S_t > K, \\ 0, & S_t < K, \end{cases}$$

while the transition near K becomes increasingly sharp.

Complete the Black–Scholes sign drill

Under $\mathbb{P}$,

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.$$

YOUR TURN Write, in order:

1. the market price of risk;
2. the density process;
3. the $\mathbb{Q}$-Brownian motion;
4. the $\mathbb{Q}$-dynamics of S;
5. the dynamics of $e^{-rt}S_t$.